Week 4 – Data Visualization & Capstone Project

End-to-End Retail Sales Analysis and Sales Prediction

Dataset integrity note: The Google Sheets dataset URL included in the assignment could not be accessed in this environment. This report therefore uses a reproducible synthetic retail-sales dataset, not actual company records. Replace it with the official data before making real business claims.

1. Executive Summary

The project demonstrates a complete analytics workflow across 1,499 cleaned transactions. Total sales were 979,663.40, total profit was 331,725.50, profit margin was 33.9%, and average order value was 653.54. A Random Forest regression model was evaluated on a held-out test set (MAE 54.38; RMSE 187.25; R^2 0.977).

2. Objectives and Tools

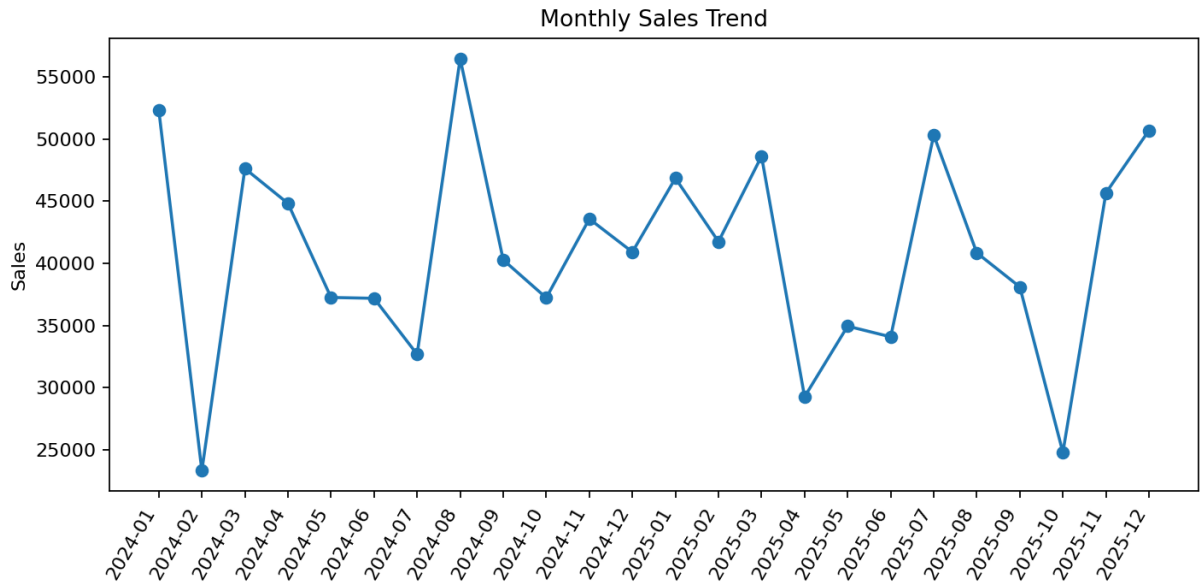
Objectives: inspect and clean transactional data; perform descriptive and visual EDA; analyze sales and profit by time, category, region and segment; train and evaluate a regression model; and prepare Power BI-ready data and measures. Tools: Python, Pandas, NumPy, Matplotlib, scikit-learn, Excel and Power BI Desktop.

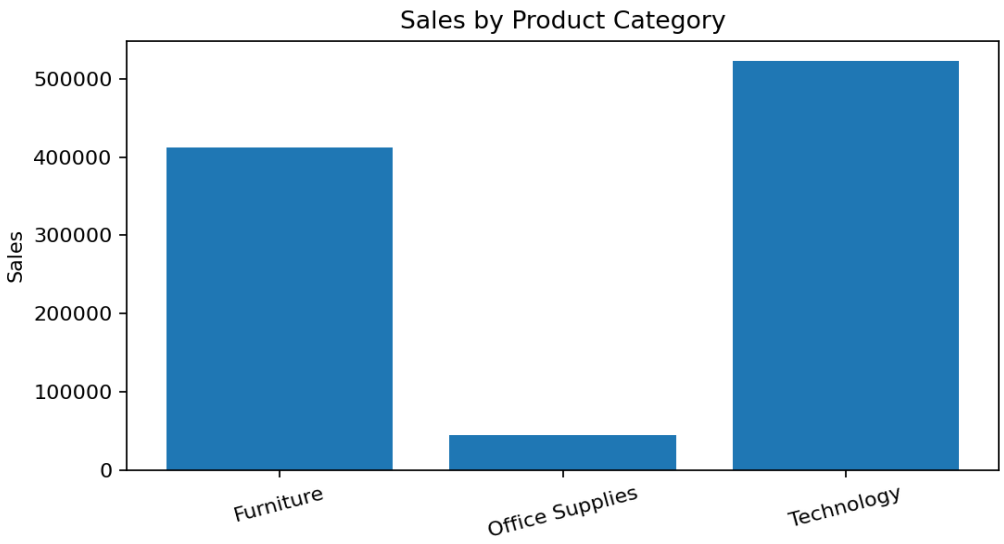
3. Data Cleaning and Preparation

The raw file contains 1,500 rows. Cleaning removed duplicate Order_ID records, filled missing region values with 'Unknown', imputed missing quantities with the median, converted dates and numeric fields, and derived Month and Profit_Margin. The resulting analysis dataset contains 1,499 rows. The raw and cleaned records are retained in the Excel workbook for auditability.

4. Exploratory Data Analysis

Technology was the highest-revenue category (522,266.63). West led regional revenue (284,159.01), while West led regional profit (99,394.76). The strongest month was 2024-08 (56,459.54).

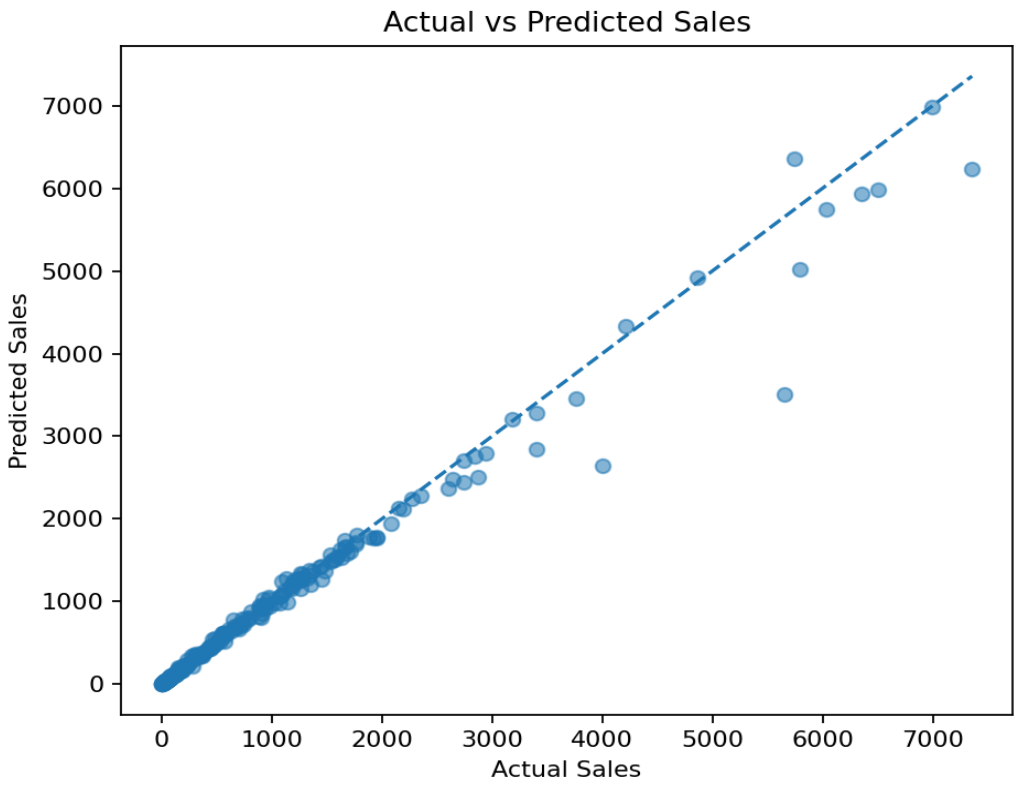



Sales by Product Category
Sales
500000
400000
300000
200000
100000
0
Furniture
Office Supplies
Technology

5. Regression Model

A Random Forest Regressor was selected because it can model non-linear relationships and handle mixed numeric and categorical predictors. Inputs were Quantity, Unit_Price, Discount, Region, Category, Sub_Category, Segment and Channel. Sales was the target. OneHotEncoder was used for categorical variables, and an 80/20 random train-test split was used.

Metric	Test result
MAE	54.38
RMSE	187.25
R^2	0.977



6. Findings and Recommendations

The analysis covers **1,499 cleaned transactions** from a reproducible sample retail dataset spanning 2024-01-01 to 2025-12-30. The data is synthetic and is intended to demonstrate the end-to-end analytics workflow; it is not a record of an actual company.

- **Revenue:** total sales were 979,663.40, with total profit of 331,725.50 and an overall profit margin of 33.9%.
- **Order value:** average sales per order were 653.54.
- **Category:** Technology generated the highest sales (522,266.63). Technology had the lowest category-level profit margin (33.4%), indicating that revenue alone should not drive assortment decisions.
- **Region:** West led revenue (284,159.01); West generated the highest profit (99,394.76). Review both revenue and margin when allocating sales resources.
- **Monthly performance:** the strongest month was 2024-08, with sales of 56,459.54.
- **Prediction:** the Random Forest regression model achieved MAE 54.38, RMSE 187.25, and R^2 0.977 on a held-out 20% test set.

Recommendations

1. Protect profitable category and region combinations; evaluate profit margin alongside sales before increasing promotion or inventory.
2. Investigate discount-heavy transactions and monitor whether discounts increase volume enough to offset margin erosion.
3. Review monthly trends for seasonality and plan inventory and staffing ahead of stronger periods.
4. Use the regression model as a planning aid, not an automated pricing or financial decision system. Validate on fresh, real transaction data before operational use.
5. Replace the sample data with the official assignment Google Sheet before presenting the findings as business facts.

Limitations

The dataset is synthetic, so the numerical insights are illustrative only. Sales is mathematically determined by quantity, unit price and discount in the generated data, which makes the prediction task demonstrative rather than a proof of real-world forecasting ability. A time-based validation split and external drivers (campaigns, stock availability, seasonality) should be considered with genuine data.

7. Power BI Dashboard Design

The supplied dashboard guide and DAX measures support a one-page executive dashboard with KPI cards, monthly sales trend, category and regional comparisons, segment mix, discount-profit scatter plot, and slicers for date, region, category, segment and channel. Import retail_sales_cleaned.csv into Power BI Desktop and save the native report as a .pbix file.

8. Deliverables and Reproducibility

The submission ZIP includes raw and cleaned CSVs, a runnable Python script, a Jupyter notebook, an Excel workbook with summary sheets and charts, Power BI DAX measures, a theme JSON, and dashboard instructions. Run analysis.py from the extracted folder after installing the dependencies listed in its header.

9. Conclusion

This capstone demonstrates the full path from data quality checks to EDA, regression evaluation, stakeholder communication and BI dashboard planning. The key next step is to rerun the same workflow against the official assignment dataset and refresh the Power BI report before final submission.